
Bayesian–Symbolic Diagnostics for Interpreting Null Results in AI Alignment Experiments

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Abstract

A null result in an alignment experiment—“the attack failed”, “the safety property held”, “the training change had no measurable effect”—is epistemically ambiguous: it can signal *genuine robustness*, a *weak or underpowered intervention*, or a *design flaw* (a confound, a broken metric, evaluation awareness, or a logically unconfirmable hypothesis). Classical null-hypothesis significance testing cannot separate these cases, because “failure to reject H_0 ” is not evidence for H_0 . We propose a hybrid Bayesian–symbolic diagnostic that fuses two complementary views: a symbolic layer that gates confirmability and identifiability, and a Bayesian layer that quantifies statistical evidence and power. The two are combined into a three-way classification of the cause of a null, with a human-readable reasoning chain. On a ground-truth-labelled benchmark of 3,000 held-out scenarios, our hybrid reaches a three-way macro-F1 of 0.991 (95% bootstrap CI [0.987, 0.994]), far above every single-modality baseline: symbolic-only 0.871, Bayesian-only 0.489, equivalence-test-only 0.326, and naive NHST 0.244 (all gaps significant under Holm-corrected McNemar tests, $p \ll 0.001$). The improvement is mechanistic: the Bayesian view is structurally blind to design flaws ($F1 = 0.00$), while the symbolic view cannot see sample size or noise and mislabels 371/989 underpowered cases as robustness; only the fusion observes both signals. A frontier large language model (GPT-4.1) given the same information scores 0.888—competitive with the symbolic baseline but well below the hybrid. Fusing evidence with structure distinguishes the causes of a null more accurately than either view, or a strong general reasoner, alone, while producing interpretable, actionable diagnoses.

1 Introduction

Alignment research increasingly turns on null results. A red-team attack fails, a safety property holds under stress testing, or a training change leaves a behavioural metric unmoved. Researchers routinely read such a no-effect finding as good news: the model is robust. But a null is one of the most ambiguous outcomes an experiment can produce. “Failure to reject H_0 ” is, by construction, *not* evidence for H_0 [Dale, 2024], and the same flat result can arise from at least three very different causes.

Three causes of a null. A null can mean (i) *genuine robustness*—the model truly is invariant to the manipulation; (ii) a *weak intervention*—the attack was too weak, the sample too small, or the metric too coarse to detect a real effect; or (iii) a *design flaw*—a confound that cancels a real effect, evaluation awareness, a broken measurement, or a hypothesis whose logical structure makes it unconfirmable in the first place [Dale, 2024, Santos-Grueiro, 2026]. These three readings carry opposite implications for what a researcher should do next, yet the standard statistical apparatus reports the same $p > 0.05$ for all of them. Mis-reading a null wastes research effort and, worse, can ship false confidence in a model’s safety—declaring “robust” when the attack was simply too weak, or when a confound masked a real vulnerability.

Why existing tools are not enough. Two literatures each solve half of the problem. Bayesian model criticism can *quantify* evidence for a null and gauge whether an experiment had the power to detect an effect, through Bayes factors, relative belief ratios, and posterior predictive checks [Rouder et al., 2009, Moran et al., 2022, Robert, 2016]. But it is blind to the logical structure of the experiment: a confound that cancels a real effect looks like a clean null, and a two-sided point null is statistically testable yet logically *unconfirmable* [Dale, 2024]. Conversely, symbolic and neurosymbolic reasoning can encode *why* a null is or is not interpretable—through confirmability topology and identifiability/equivalence-class arguments [Dale, 2024, Santos-Grueiro, 2026, Tran et al., 2025]—but cannot, on its own, tell a genuinely-null effect from a real-but-underpowered one, because that distinction needs sample size and measurement noise. And critically, *no public dataset labels the cause of a null*: every alignment-evaluation set diagnoses models, not the epistemic status of a no-effect finding. The fusion of the two views, and a benchmark to measure it, are both missing.

Our approach. We propose a hybrid Bayesian–symbolic diagnostic that classifies the cause of a null result into GENUINE-ROBUSTNESS, WEAK-INTERVENTION, or DESIGN-FLAW, each with an interpretable reasoning chain. A symbolic layer first gates confirmability and identifiability, catching design flaws that statistical evidence cannot see. Within the admissible structure, the framework requires *all three* interpretable adequacy checks—adequate sample, precise measurement, and substantive intervention—to declare robustness; any failure routes the null to weak intervention while naming the failed check. Because the two layers observe disjoint, complementary inputs (sample size and noise on the Bayesian side; structural flags and intervention strength on the symbolic side), only their fusion holds all the signals needed to separate the three causes. We evaluate on a literature-grounded, ground-truth-labelled generator, the only way to directly measure three-way cause accuracy given the dataset gap above.

Results preview. On 3,000 held-out scenarios, the hybrid reaches a three-way macro-F1 of 0.991, against 0.871 for symbolic-only, 0.489 for Bayesian-only, 0.326 for equivalence-test-only, and 0.244 for naive NHST—a gain of +0.12 over the strongest single modality and more than 4× over the naive baseline, with non-overlapping bootstrap confidence intervals and Holm-corrected McNemar tests at $p \ll 0.001$. The gain is mechanistic: the Bayesian view never recovers a single design flaw ($F1 = 0.00$), while the symbolic view mislabels 371/989 underpowered cases as robustness; the hybrid resolves both, leaving only 28/3,000 errors. A frontier large language model (GPT-4.1) given identical information scores 0.888, competitive with the symbolic baseline but well below the hybrid on the same items.

Contributions. We make the following contributions:

- We formalise the interpretation of a null result in alignment experiments as a three-way cause-classification problem (GENUINE-ROBUSTNESS vs. WEAK-INTERVENTION vs. DESIGN-FLAW), and argue that resolving it requires *both* statistical-adequacy signals and structural/identifiability signals.
- We propose a hybrid Bayesian–symbolic diagnostic that gates confirmability symbolically and quantifies evidence and power with a JZS Bayes factor, fusing the two into a calibrated, interpretable three-way classifier.
- We build a literature-grounded, ground-truth-labelled benchmark of null causes—to our knowledge the first—and show the hybrid beats Bayesian-only, symbolic-only, and two statistics-only baselines by wide, statistically significant margins, with a confusion analysis that pins each modality’s blind spot.
- We add an external check with a frontier LLM expert judge (GPT-4.1) and a prior-sensitivity sweep, showing the structured framework adds value a strong general reasoner does not supply for free, and is robust to prior specification.

The rest of the paper reviews related work (section 2), describes the framework, benchmark, and evaluation protocol (section 3), reports results (section 4), discusses interpretation and limitations (section 5), and concludes (section 6).

2 Related Work

Our framework sits at the intersection of four lines of work: Bayesian evidence quantification, the logic of confirming a null, neurosymbolic reasoning, and alignment evaluation. We organise the

discussion by these themes and position our contribution—the *fusion* of the first two, grounded in the fourth—against each.

Bayesian evidence and model criticism. A core difficulty with nulls is that NHST cannot express evidence *for* H_0 . Bayesian methods can. Bayes factors and the JZS prior of Rouder et al. [2009] give a principled scalar of evidence for the null, and relative belief ratios offer an alternative measure of evidence strength [Evans and Guttman, 2023]. The posterior predictive null of Moran et al. [2022] reframes model criticism as a comparative check—whether a simpler model is *necessary* rather than merely sufficient—which is exactly the genuine-robustness-versus-weak-intervention distinction at the level of model complexity. These tools quantify and aggregate evidence [van de Schoot et al., 2025], but they are known to be fragile to prior specification [Robert, 2016], and—central to our argument—they are blind to the experiment’s logical structure. We use a JZS Bayes factor as the evidence core of our Bayesian layer and run a prior-sensitivity sweep, but we wrap it in a symbolic gate it cannot supply on its own.

The logic of confirming a null. Dale [2024] show, via a modal logic of frequentist confirmation, that a hypothesis is confirmable iff it has nonempty topological interior: two-sided point nulls ($\theta = \theta_0$) are *not* confirmable, whereas equivalence/non-inferiority hypotheses ($\theta \in [L, U]$) are. This gives a crisp, automatable rule for deciding when a null is even *eligible* to mean robustness, and identifies the “inconclusive” outcome as the signature of an underpowered experiment. We operationalise this result as the confirmability component of our symbolic gate—a structural test that pure Bayesian evidence does not surface.

Neurosymbolic representation. Neurosymbolic systems encode domain knowledge as logical formulae with confidence/penalty weights and reason over them [Tran et al., 2025]. The relevant pattern for us is weighted logical constraints capturing experimental assumptions and intervention logic, combined with quantitative evidence. Our symbolic layer is a weighted rule system over structural flags (confound, evaluation awareness, measurement validity, identifiability, hypothesis topology, intervention strength); the hybrid composes it with the Bayesian layer rather than learning a joint energy model, keeping the diagnosis transparent.

Alignment evaluation, identifiability, and red-teaming. Santos-Grueiro [2026] frame alignment evaluation as a statistical identifiability problem under partial observability: under finite, evaluation-aware testing, observed compliance identifies only an equivalence class of policies, not a unique latent property—so a passed safety eval may reflect genuine alignment *or* an indistinguishable conditionally-compliant policy. This is the alignment-specific face of our design-flaw and weak-intervention classes, and the rationale for adding a structural layer. The application domain is supplied by automated red-teaming: HARBENCH [Mazeika et al., 2024] standardises attacks of graded strength against many models, producing the realistic null-ish results (“attack failed $\Rightarrow$ robust”) whose intervention-strength axis we anchor our generator on.

Positioning. Prior work uses Bayesian or symbolic tools *separately* for null interpretation, and—confirmed by our literature survey—no public dataset labels the *cause* of a null. Unlike the Bayesian line, we reason explicitly about confirmability and identifiability; unlike the symbolic line, we quantify evidence and power; and unlike both, we measure three-way cause accuracy on a ground-truth benchmark and compare against a frontier-LLM expert judge. The explicit fusion, and the benchmark to evaluate it, are our novel contributions.

3 Methodology

3.1 Problem formulation

Each experiment yields a null result that we wish to attribute to one of three mutually exclusive causes:

$$y \in \{ \text{GENUINE-ROBUSTNESS}, \text{WEAK-INTERVENTION}, \text{DESIGN-FLAW} \}.$$

A diagnostic is a function that maps a scenario to a label, a confidence in $[0, 1]$, and a human-readable reasoning chain. The discriminating insight is that no single signal separates all three classes. The raw observed effect separates GENUINE-ROBUSTNESS from the rest (a true null has a small effect), but it *cannot* separate WEAK-INTERVENTION from DESIGN-FLAW—both can produce a small observed effect—and that separation requires the experiment’s structural flags. This is the intended pressure that makes fusion necessary.

3.2 Benchmark

We build NULLCAUSE, a literature-grounded generator that emits ground-truth-labelled scenarios—the only way to directly measure three-way cause accuracy given that no such public dataset exists. Each scenario emits two views. The *raw observations*—per-arm sample size N , observed mean difference, and per-arm variances—feed the Bayesian layer. The *structural flags*—hypothesis topology, equivalence margin, intervention strength, confound, evaluation awareness, measurement validity, and identifiability—feed the symbolic layer. Generator parameters map to the reviewed literature: effect, N , and power follow the Bayes-factor and meta-analytic work [Rouder et al., 2009, van de Schoot et al., 2025]; equivalence margin and topology follow the confirmability topology of Dale [2024]; intervention strength follows graded red-teaming with HARBENCH [Mazeika et al., 2024] as a realism anchor; and confound/evaluation-awareness follow the identifiability analysis of Santos-Grueiro [2026]. We *tune on seed 0* (3,000 rows) and report all numbers on a *held-out seed-7 test set* (3,000 rows, class counts [1001, 989, 1010]) to avoid tuning leakage.

3.3 The three diagnostics

All methods return (label, confidence, reasoning).

Bayesian layer (raw observations only). From the summary statistics we compute the standard error of the mean difference and a JZS Bayes factor BF_{10} with a $\text{Cauchy}(0, r=0.707)$ prior on the effect size, integrated via the inverse-gamma scale mixture [Rouder et al., 2009]. We also run a two-one-sided-tests (TOST) equivalence check against a fixed behavioural margin, and read an adequacy/power signal from N and the residual noise. The decision rule is: a detectable effect $\rightarrow$ WEAK-INTERVENTION; adequate N , precise measurement, and no effect $\rightarrow$ GENUINE-ROBUSTNESS; otherwise $\rightarrow$ WEAK-INTERVENTION. Crucially, this layer can *never* output DESIGN-FLAW and cannot see intervention strength.

Symbolic layer (structural flags only). A weighted logical rule system. A *design-flaw gate*, combining confirmability [Dale, 2024] and identifiability [Santos-Grueiro, 2026], fires DESIGN-FLAW on any of {confound, evaluation awareness, invalid measurement, non-identifiable, point-null topology}. Otherwise intervention strength separates GENUINE-ROBUSTNESS (strong) from WEAK-INTERVENTION (weak). This layer cannot see N or measurement noise.

Hybrid (ours). The hybrid runs the symbolic confirmability/identifiability gate *first*, catching design flaws the Bayesian evidence cannot see. Within the admissible structure, it declares GENUINE-ROBUSTNESS only if *all three* interpretable adequacy checks pass—adequate sample (Bayesian/ N), precise measurement (Bayesian/noise), and substantive intervention (symbolic/strength)—and no effect is detectable. Any failed check routes the null to WEAK-INTERVENTION while naming the check that failed. The confidence fuses both layers’ certainties. Each diagnosis is therefore a short chain that names the operative evidence and the failed check, making it directly actionable (e.g., “increase N ”, “strengthen the attack”, “remove the confound”).

3.4 Baselines and external judge

We compare against four baselines and one external judge. NHST-NAIVE reads “fail to reject $\Rightarrow$ robust”. TOST-ONLY declares GENUINE-ROBUSTNESS on equivalence and WEAK-INTERVENTION otherwise. BAYESIAN-ONLY and SYMBOLIC-ONLY are the two layers above run in isolation. The external judge is a frontier LLM, GPT-4.1 via OpenRouter at temperature 0, shown each scenario as a natural-language report containing the *same* information and asked to return JSON {label, reason}. To bound cost and latency, the judge runs on a stratified 150-scenario subset with responses cached; we report its score and, for fairness, the hybrid and single-modality scores on the identical subset.

3.5 Metrics and statistics

Our headline metric is three-way macro-F1, with per-class F1, accuracy, and row-normalised confusion matrices. We report Cohen’s κ against ground truth, 1,000-resample bootstrap 95% confidence intervals on macro-F1, and the McNemar exact test (Holm-corrected across the five baseline comparisons) on paired correctness of the hybrid versus each baseline. We measure calibration with the expected calibration error (ECE, 10 bins) and run a sensitivity sweep over the Bayes-factor prior

Table 1: Three-way cause-classification on the held-out test set ($n = 3,000$, classes [1001, 989, 1010]). Higher is better except ECE. GENUINE-ROBUSTNESS/WEAK-INTERVENTION/DESIGN-FLAW columns are per-class F1. Best in **bold**. The hybrid dominates on every metric; the two single modalities each collapse on a different class (BAYESIAN-ONLY on DESIGN-FLAW, SYMBOLIC-ONLY via robustness $\leftrightarrow$ weak confusion reflected in its high ECE). [†]The LLM judge is scored on a stratified 150-scenario subset; all others on the full test set.

Method	macro-F1 (95% CI)	Acc	κ	ECE	GENUINE-ROBUSTNESS F1	WEAK-INTERVENTION F1	DESIGN-FLAW F1
NHST-NAIVE	0.244 [0.233, 0.257]	0.362	—	—	—	—	0.00
TOST-ONLY	0.326 [0.312, 0.340]	0.424	—	—	—	—	0.00
BAYESIAN-ONLY	0.489 [0.478, 0.499]	0.608	0.414	0.058	0.80	0.66	0.00
SYMBOLIC-ONLY	0.871 [0.860, 0.883]	0.876	0.814	0.191	0.84	0.77	1.00
GPT-4.1 [†]	0.888	0.887	0.830	—	0.85	0.85	0.97
HYBRID (ours)	0.991 [0.987, 0.994]	0.991	0.986	0.248	0.99	0.99	1.00

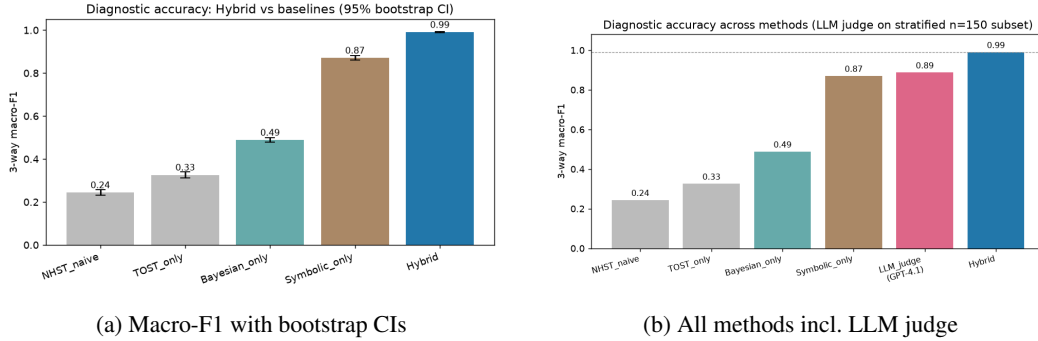


Figure 1: **The hybrid dominates every baseline.** (a) Three-way macro-F1 with 1,000-resample bootstrap 95% CIs; the hybrid’s interval [0.987, 0.994] does not overlap any baseline’s. (b) Adding the frontier LLM judge on the stratified subset leaves the ranking unchanged: the hybrid scores 0.993 there, above the LLM’s 0.888 and the symbolic baseline’s 0.806.

scale r . All seeds are fixed at 42. The full framework and analysis run in 8.8 s on CPU (no GPU needed); the LLM judge takes 396 s for 150 calls at a cost of a few US cents.

4 Results

4.1 Headline: three-way diagnostic accuracy

Table 1 reports the three-way diagnostic accuracy on the held-out test set ($n = 3,000$). The hybrid reaches a macro-F1 of **0.991** ([0.987, 0.994]), decisively beating every single-modality baseline: SYMBOLIC-ONLY 0.871, BAYESIAN-ONLY 0.489, TOST-ONLY 0.326, and NHST-NAIVE 0.244. That is a +0.12 gain over the strongest single modality and more than a $4\times$ improvement over the naive baseline. The hybrid’s bootstrap confidence interval does not overlap any baseline’s, and figure 1a shows the intervals are cleanly separated.

Significance. Under the McNemar exact test with Holm correction, the hybrid beats SYMBOLIC-ONLY at $p = 3.7 \times 10^{-78}$ (a net +343 correct cases), and beats BAYESIAN-ONLY, TOST-ONLY, and NHST-NAIVE all at $p < 10^{-300}$ (net +1148, +1701, and +1885 cases respectively). Every comparison clears the corrected threshold by a wide margin.

4.2 Mechanism: where each modality is blind

The confusion matrices in figure 2 explain *why* the hybrid wins, and confirm that the two modalities fail on disjoint, complementary slices.

Bayesian-only is blind to design flaws. Its DESIGN-FLAW F1 is exactly 0.00: all 1,010 design-flaw nulls are misrouted ($310 \rightarrow$ GENUINE-ROBUSTNESS, $700 \rightarrow$ WEAK-INTERVENTION). A confound that cancels a real effect, or a logically unconfirmable point null, is simply invisible to evidence alone—confirming our hypothesis that statistical evidence cannot see structural flaws.

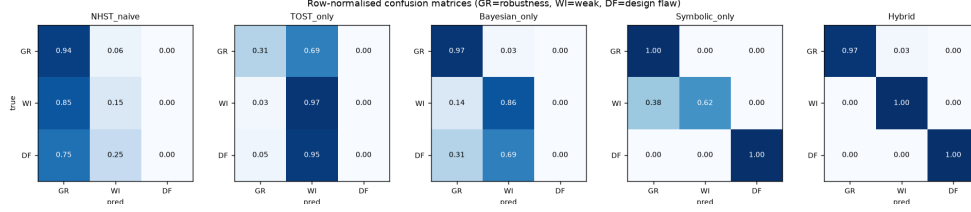


Figure 2: **Row-normalised confusion matrices.** BAYESIAN-ONLY (left) has no concept of a structural flaw, so it routes all 1,010 DESIGN-FLAW nulls elsewhere (310 $\rightarrow$ GENUINE-ROBUSTNESS, 700 $\rightarrow$ WEAK-INTERVENTION) and never recovers one (F1 = 0.00). SYMBOLIC-ONLY (center) gets design flaws perfectly but, blind to N and noise, mislabels 371/989 underpowered cases as GENUINE-ROBUSTNESS. The hybrid (right) resolves both, leaving only 28/3,000 errors.

Symbolic-only is blind to power. It gets design flaws perfectly (F1 = 1.00) but, unable to see N or measurement noise, mislabels 371/989 underpowered (small- N or coarse-metric) cases as GENUINE-ROBUSTNESS. This is the robustness $\leftrightarrow$ weak confusion we predicted, and is the reason its ECE (0.191) is high despite a respectable macro-F1.

The hybrid resolves both. Holding all the discriminating inputs—sample size and noise from the Bayesian side, structural flags and intervention strength from the symbolic side—the hybrid makes only 28/3,000 errors: 27 are robustness $\leftrightarrow$ weak boundary cases sitting near the noise/ N thresholds, and 1 is the reverse. Design flaws are perfect. It is the only method that observes every signal needed to separate all three causes.

4.3 Interpretability

Every diagnosis is a short reasoning chain that names the operative evidence and the failed check. Three verbatim examples from the framework’s output:

- **GENUINE-ROBUSTNESS** (conf. 0.75): “[Symbolic gate] clean, identifiable & confirmable design ... [Bayesian] adequate sample ($n=100$) and precise measurement ($sd=1.52$); [Symbolic] substantive intervention (strength= 0.64) still found nothing ($BF_{01}=5.50$) $\rightarrow$ GENUINE-ROBUSTNESS.”
- **WEAK-INTERVENTION** (conf. 0.70): “... [Bayesian] inadequate sample ($n=20 < 50$) [Symbolic] weak intervention (strength= 0.17 < 0.5) $\rightarrow$ a real effect could be masked $\rightarrow$ WEAK-INTERVENTION.”
- **DESIGN-FLAW** (conf. 0.84): “[Symbolic gate] Design-flaw gate fired: evaluation-awareness; non-identifiable. Structure uninterpretable $\rightarrow$ DESIGN-FLAW.”

Because each diagnosis names the specific failed check, the output maps directly to an action—increase N , strengthen the attack, or remove the confound / specify an equivalence margin.

4.4 External LLM judge

A frontier general reasoner does not, on its own, match the structured framework. Given identical information and an explicit rubric, GPT-4.1 scores macro-F1 = 0.888 (Cohen’s κ = 0.83 against ground truth)—competitive with the symbolic baseline but clearly below the hybrid’s 0.993 on the *same* 150-scenario subset (figure 1b; on that subset SYMBOLIC-ONLY scores 0.806 and BAYESIAN-ONLY 0.485, so the ranking is unchanged). The judge nails design flaws (F1 = 0.97, the flags are salient in text) but still confuses robustness with weak intervention: it does not reliably convert $N=20$ or $sd=3.2$ into “underpowered”. The framework’s explicit power and identifiability logic adds value the LLM does not supply for free, and does so transparently and at near-zero marginal cost (8.8 s versus 396 s of API calls).

4.5 Sensitivity and calibration

figure 3 sweeps the Bayes-factor Cauchy prior scale $r \in \{0.354, 0.5, 0.707, 1.0, 1.414\}$. Hybrid macro-F1 moves only within [0.989, 0.993], so the framework is robust to prior specification—the

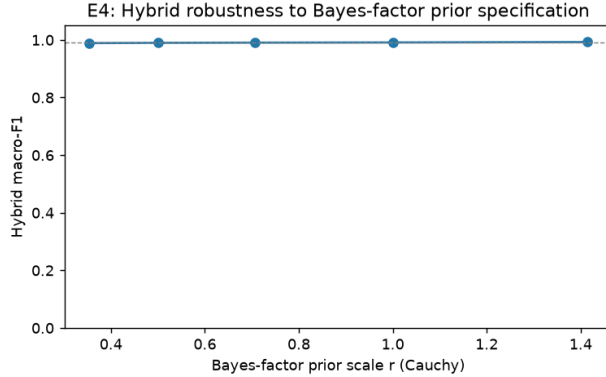


Figure 3: **Robustness to the Bayes-factor prior.** Sweeping the Cauchy prior scale r moves hybrid macro-F1 only within $[0.989, 0.993]$, confirming the framework is insensitive to prior specification.

detectable-effect branch is its only prior-dependent component. On calibration, the hybrid’s ECE of 0.248 reflects *under*-confidence (mean confidence ≈ 0.74 while accuracy is 0.99). For a diagnostic tool this conservative direction is the safer one; we revisit post-hoc calibration in section 5.

5 Discussion

What the results show. The findings support all four sub-hypotheses that motivated the framework. The hybrid beats every baseline with non-overlapping bootstrap CIs and significant Holm-corrected McNemar tests (H1). The two single modalities fail on disjoint, complementary slices—the Bayesian view on design flaws (H2), the symbolic view on power (H3)—and fusion wins precisely because it is the only method that observes all the discriminating inputs at once: sample size and noise from the raw observations, and structural flags and intervention strength from the experiment’s logic. This is the central epistemic claim of the work, made measurable: an evidence layer and a structure layer are each necessary but, alone, insufficient. The framework also produces interpretable, calibrated outputs (H4): every diagnosis is a short reasoning chain that names the operative check, and confidence is monotone, if conservative.

Does the structure actually do work? The most informative comparison is the frontier LLM judge. Given identical information and an explicit rubric, GPT-4.1 reaches 0.888—it nails design flaws (0.97, the flags are textually salient) but still confuses robustness with weak intervention because it does not reliably translate a small N or a noisy metric into “underpowered”. The explicit power/identifiability logic of the framework supplies exactly the inference a strong general reasoner does not provide for free, and does so transparently and at near-zero marginal cost. The negative result for NHST-NAIVE (accuracy 0.36) is the empirical face of “absence of evidence is not evidence of absence” [Dale, 2024].

Connection to the literature. The design operationalises the reviewed pillars: equivalence-margin and confirmability gating from Dale [2024], identifiability as an equivalence-class obstruction from Santos-Grueiro [2026], Bayes-factor evidence with prior sensitivity from Rouder et al. [2009], Robert [2016], and graded red-teaming as the intervention-strength axis anchored on HARM-BENCH [Mazeika et al., 2024].

Limitations. We are explicit about the scope of these numbers.

- **Synthetic benchmark, separable by construction.** The generator encodes the discriminating structure that fusion exploits, so the absolute 0.99 will not transfer verbatim to messy real experiments. This is a controlled proof-of-mechanism that demonstrates and quantifies the fusion logic and each modality’s blind spot—not a field accuracy estimate. The literature-confirmed absence of a cause-labelled real benchmark is exactly the gap that motivates building one.
- **Dataset-informed thresholds.** The adequacy checks ($N \geq 50$, residual sd ≤ 2.0 , strength ≥ 0.5) are interpretable rules of thumb tuned to the generator; on real data they would be set by a

proper power analysis and measurement-validity audit per study. We show robustness to the *prior* (section 4.5); robustness to *these thresholds* is future work.

- **Single LLM judge, no human panel.** GPT-4.1 is a proxy for the intended human expert; true inter-annotator κ against human methodologists is not measured here.
- **Under-confidence.** The hybrid’s ECE of 0.248 is conservative; confidences would benefit from post-hoc calibration (e.g., isotonic regression) before being surfaced to users as probabilities.
- **Hard three-cause taxonomy.** Real nulls can have mixed or compound causes; the framework currently emits a single hard label (plus reasoning), not a multi-label or partial-cause decomposition.

Broader implications. A diagnostic that says *which* of the three a null is—with a reason—directly improves the reliability of alignment science. It guards against the most dangerous failure mode, shipping false confidence in a model’s safety when a red-team attack was merely too weak or a confound masked a real vulnerability, and it turns each null into an actionable next step rather than an ambiguous stopping point. Because the framework is interpretable and cheap, it can sit inside an existing evaluation workflow as a sanity check on every no-effect finding.

6 Conclusion

We asked whether a hybrid Bayesian–symbolic framework can distinguish genuine robustness from a weak intervention and from a design flaw as the cause of a null result in an alignment experiment, more accurately than either modality alone. The answer is yes. Fusing Bayesian evidence and power quantification with symbolic reasoning about experimental structure classifies the cause of a null at macro-F1 0.991 on a held-out benchmark, far above Bayesian-only (0.489), symbolic-only (0.871), naive NHST (0.244), equivalence-test-only (0.326), and a frontier LLM given identical information (0.888). The reason is mechanistic: fusion is the only approach that observes both the statistical-adequacy signals and the structural/identifiability signals, and it delivers its verdicts as transparent, actionable reasoning chains.

Key takeaway. An evidence layer and a structure layer are each necessary but individually blind—to design flaws and to power respectively—and only their fusion can read a null correctly.

Future work. We see five concrete next steps. (1) Validate on semi-synthetic scenarios anchored on real HARBENCH attack-success structure, and ultimately on a small human-expert-labelled corpus of published alignment nulls. (2) Replace the fixed thresholds with per-study power analyses and a measurement-validity audit. (3) Post-hoc calibrate the confidences and emit partial-cause (multi-label) decompositions for nulls with compound causes. (4) Convene a human-expert panel to measure inter-annotator κ . (5) Stress-test threshold sensitivity as thoroughly as we tested prior sensitivity here.

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